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On the Impact of Deployment Geometry and Radio Range on Random Walk–Based Source Location Privacy in WSNs

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ABSTRACT This research aims to understand the impact of the network area and shape on the performance metrics used in evaluating the random walk-based source location privacy (SLP) techniques developed for WSNs. In this context, the impact of circular and square network models for different network areas, node densities, and radio ranges of sensor nodes on the performance of three popular SLP techniques is investigated. The effectiveness is assessed using performance measures from the body of available literature. It has been found that square deployment performs better for sector-based SLP protocols when network area, node density, and node radio range are held constant. However, circular networks perform better for the same protocol when the radio range is varied beyond a certain threshold while holding all other parameters constant. The trend, however, changes under comparable network settings for the non-sector-based random walk protocols considered in the current work. The results reveal a strong link between deployment geometry and routing logic, demonstrating that sector-based SLP protocols are primarily geometry-sensitive, whereas non-sector-based protocols are more sensitive to radio-range variations. Furthermore, a radio-range threshold is identified beyond which privacy and efficiency trends reverse or saturate, indicating diminishing returns from increasing transmission range. These findings provide practical design guidelines for geometry-aware deployment of privacy-preserving WSNs.

INDEX TERMS IoT, Performance Effectiveness, Network Shape and Size, Radio Range, Source Location Privacy, WSN.

I. INTRODUCTION

DURING the last couple of decades, Wireless Sensor Networks (WSNs) have found favor for the deployment of several applications and hence have been investigated extensively across the globe. In more recent times, variants of WSN such as the Internet of Things (IoT) have been explored with a higher priority because of the availability of low-cost, short-range sensors and their ease of deployment. WSN systems are typically designed to detect, track, and transmit real-time data from specified monitoring settings to back-end systems for additional processing and analysis. WSNs are used in numerous applications ranging from health care, agriculture, home automation, automotive networks, and Industry 4.0 to mention a few [1]–[6]. Furthermore, WSNs have also been used in landslide detection monitoring, underwater applications [7], [8], military, and wildlife protection [9]. In the context of wildlife protection, WSNs are used to track endangered species and provide updates on the whereabouts and health status of these animals [10]. Since networks for such applications are deployed in remote locations with scarce resources, substantial challenges in regard to network security, deployment, coverage, data collection, and Source

Location Privacy (SLP) [11]–[14] are often observed.

In this light, SLP for endangered species has garnered a lot of attention from the research community across the globe and is also the theme of the current study. As briefly mentioned earlier, the sensor nodes in a WSN consist of simple electronics with low processing capability which makes them easy to deploy while making them vulnerable to unwanted access making SLP a critical responsibility [15]. Basically, SLP deals with safeguarding the location of endangered animals (or events/assets in general) from attackers and intruders.

To understand the importance of SLP in WSN applications, let us consider a Panda-Hunter game model proposed in [16]. In this scenario, the pandas are thought to live in a monitoring area with a huge number of sensor nodes that are dispersed randomly and uniformly. When a panda is detected, the sensor node that detected the panda becomes the source and sends the monitored message in a multi-hop way to the Base Station (BS). However, the hunter/poacher whom we term the attacker or the adversary is all set for poaching using Information and Communication Technologies (ICT). The attacker begins its journey from the BS by analyzing the received packet to identify the immediate sender node by *backtracking*. The hunter repeats the procedure until it reaches the source node (SN) and finally captures the panda. Therefore, it is important to conceal the position (i.e., ensure location privacy) of the node that identifies an asset in its radio range. To mitigate such issues, several solutions have been proposed in literature for preserving SLP in WSNs. The solutions have been developed by considering different types of attacker models (local, semi-global, and global attacker), network models for node deployment (random, strategic, square, and circular networks), and location concealment strategy (routing-, fake packets-, fake source-, multiple BS-based schemes).

Despite extensive research on routing-based SLP protocols, most existing studies implicitly assume that network geometry remains fixed or plays a negligible role once node density and radio range are defined. Consequently, the interaction between deployment geometry and routing logic—and its impact on privacy and efficiency metrics—remains poorly understood. This gap is critical because real-world WSN and IoT deployments are often constrained by terrain, coverage shape, and radio hardware limitations. To address this gap, we study and evaluate the influence of two deployment geometries—square and circular—on the effectiveness of three existing random walk-based SLP techniques, namely Sector-based Random-walk Routing (SRR), Phantom Routing Based Pure Random Walk (PRPRW), and Phantom Routing Based L-Path Pure Random Walk (PRLPRW), under controlled network conditions where two of the three parameters—network area, node density, and sensor nodes' radio range—are held constant.

The major contributions of the current study can be enumerated as follows:

- We provide the first systematic quantification of the interaction between deployment geometry (square vs. circular) and random walk–based SLP protocols under controlled network area, node density, and radio range.
- We demonstrate that sector-based SLP protocols exhibit strong sensitivity to deployment geometry, whereas non-sector-based protocols are predominantly influenced by radio-range variations.
- We identify a radio-range threshold beyond which improvements in privacy and efficiency metrics saturate or reverse, highlighting a critical design tipping point.
- Based on extensive simulation results, we derive actionable design insights for selecting deployment geometry and protocol configurations in privacy-sensitive WSN applications.

The rest of the paper is organized as follows: In sec. II, we give an overview of the existing SLP techniques. These SLP techniques have used either square or circular deployment models. In sec. III and sec. IV, we explain the network and attacker models considered in the work. We then describe the three SLP techniques used as case study protocols for this research in sec. V. The simulation settings and evaluation metrics are presented in sec. VI and while the results from the current investigation are reported in sec. VII respectively. In sec. VIII we provide the guidelines on the networks deployment. We finally infer the current work in sec. IX.

II. LITERATURE REVIEW

In this section, we present an overview of the existing state-of-the-art SLP protocols in the literature. The review of the protocols is based on the shape of the network deployment—square and circle.

A. SLP SOLUTIONS WITH SQUARE DEPLOYMENT MODEL

The idea of a square deployment model for WSN was initially introduced and elaborated by Jhumka et al. in [17]–[19]. The work in [17] focused on designing dynamic online parameter estimation for protecting source privacy in WSN where the primary focus was on understanding the trade-off between a temporary fake source period, a temporary fake source duration, and a permanent fake source period. In contrast, the objective of the work in [18] was to provide a decision-theoretic procedure for selecting an appropriate SLP routing algorithm based on the application needs and the network under investigation. Finally, a multi-objective optimization problem was suggested in [19] where SLP, schedule latency, and attacker distance were considered as the optimization parameters. Furthermore, a scheduling algorithm was proposed for the generation of fake packets based on the optimization of the parameters and the generic algorithm. In comparison three levels of security for SLP in WSNs were proposed in [20] where a ring surrounding the central BS was the flooding zone for fake packets, a node in a direction away from BS served as a fake node, and an additional fake source in the appropriate quadrant of the network that added to the ambiguity of the attacker.

To improve network lifetime a dynamic multipath privacy-preserving routing (DMPPR) protocol was proposed in [21] where portions of packets were transmitted using random phantom nodes through multiple dynamic paths. The introduced distributed routing approach resulted in better utilization of nodes in the network while also improving the security of transmission through closed-loop implementation.

While the previous techniques majorly delved into single-asset scenarios, two phantom routing techniques were proposed for multi-asset scenarios in [22]. The authors proposed two phantom routing-based methods for the scenario where multiple assets co-exist in the network. In the first approach namely, phantom routing based backward random walk (PRBRW), packets were routed to the BS by using a mix of backward random walk and a greedy forwarding approach which while improving the security of transmission in terms of capture ratio and safety period had an adverse effect on network lifespan and entropy. In comparison the second strategy, called phantom routing-based L-path random walk (PRLPRW), not only improved the capture ratio and safety period but also entropy and network lifetime thereby providing a more secure routing strategy.

B. SLP SOLUTIONS WITH CIRCULAR DEPLOYMENT MODEL

Circular deployment models in the domain of WSN were seen initially in the works by Tao et al. and Anfeng et al. in [23] and [24] respectively. However, these works focused on energy-efficient routing techniques for WSNs with reliability features enabled using secret sharing schemes rather than on SLP.

Drawing inspiration from the implementations in [23], [24], the authors in [25] for the first time proposed a novel SLP solution. In this approach named SLP using Randomized Routes (SLP-R), the network was divided into circular rings with the BS at the center. In this solution, the packets were initially sent in the backward direction i.e., towards the network boundary till the predefined time to live (TTL) parameter, which is a measure of the hop count, becomes zero. Once the TTL value becomes zero, the packets are sent in either a clockwise or anticlockwise direction within an angle range of $45°$ to $180°$. After this identical depth routing (IDR) phase, the packets are sent to the BS using shortest path routing (SPR) providing improved *safety period* and *network lifetime*.

Another annular ring approach for circular networks was proposed in [26]. In contrast to the previous implementation, the current technique grouped the nodes in the network into clusters. In this approach, packets were routed from a node to the BS via intra and intermediate inter-cluster heads. To add to the entropy of routing fake packets from fake sources emulating the original SN were generated.

A third circular ring-based SLP protocol was proposed for circular networks in [27]. In this technique, the network comprised three major annular segments each of a radius one-third of the network radius. Depending on the position of the SN within these segments, the protocol adapted dynamically and performed four stages of data routing: i) Backward Random Walk (BWR), ii) Clock/anti-clockwise routing, iii) SPR to the second ring, and iv) ring walk. These four phases were designed in such a way that SLP was not dependent on the position of SN.

In contrast to the previous approaches a sector-based random walk routing (SRR) where the circular network was logically divided into sectors was proposed in [28]. In SRR protocol the data packets were routed using randomly dispersed phantom sources placed in several sectors. The developed routing strategy favored the selection of nodes away from the SN to enhance privacy. In addition, to maintain a balance between security and network lifetime, the idea of hop threshold was introduced.

Recent works as seen in [29], [30] and [31] also use the square network model and circular network model, respectively. However, the nodes deployment is random in nature unlike the previously mentioned works that use regular gird-like and cluster-based arrangement. Further, these works provide solution for the case of multi-source scenario. Similarly, the work in [32] use square network model with random deployment of nodes. However, this work provides solution for the case of multiple sources and multiple destination. Another work based on circular network with random node deployment model is proposed in [33]. To counter traffic analysis attacks, packets follow a random walk, then traverse obfuscated and angle-based paths. They finally reach the base station via a forward random walk. However, this method employs fake packets to divert the adversary. These cases are out of the scope of this work.

A few recent works as seen in [34], [35] also work on source location privacy in WSN considering square network area with random node deployment model. However, these solution consider a tree-like networks structure and clustering approach, respectively and relies on fake packet transmission approach, which needs completely a different treatment.

Post an overview of the state-of-the-art in SLP, the network models considered in the current study are elaborated in the subsequent section.

III. NETWORK MODELS

In the current study, a network model is assumed to typically comprise several sensor nodes that communicate with a central controller (BS) through multiple hops. A sensor node is equipped to receive and send data packets, and sense required information from within its radio range. It consists of a small processor, storage unit, power source, RFID reader, and transponder. We assume that communication between nodes and between nodes and BS is secure and uses some established SLP techniques to protect the source node (SN). We further assume that the detecting range and communication range of a sensor node is the same and all the nodes have the same energy capacity and radio range. The network models thus described can be broadly classified as square and

circular network models depending on the arrangement of the nodes, which are described in the following subsections.

A. SQUARE NETWORK MODEL

As illustrated in Fig. 1, in a square network model, the nodes are arranged in a regular pattern of square-shaped rings around the BS with a constant separation of $\sqrt{2}r_s/2$ units between any two adjacent nodes, where r_s is the radio range of the nodes and $\sqrt{2}r_s/2$ is the offset distance. One possible way to deploy this type of network is, to begin with, the placement of the BS at the center of the network, then place one node to its right on the x-axis with an offset distance of $\sqrt{2}r_s/2$ units between BS and the sensor node, then the next node is placed at a point on the x-axis with the same offset value and this continues till the network edge is reached. Now, with BS as the center of the network, move on the y-axis of the network and place another sensor node at a distance of $\sqrt{2}r_s/2$, and the next node at the same distance horizontally. Repeat this procedure till the network edge is reached. Once the first quadrant is completed, then do the same for the rest of the three quadrants. Such network models are also used in [21], [27], [36], [37]. Placing nodes at an offset distance of $\sqrt{2}r_s/2$ units will ensure that internal nodes in the network will have eight neighbors and this is essential for random walk-based routing schemes.

B. CIRCULAR NETWORK MODEL

The circular deployment model is illustrated in Fig. 2 the nodes are arranged in regular patterns forming circular rings with the BS at the center. As can be observed each concentric ring in the network is separated by a distance of $\sqrt{2}r_s/2$ units—radio range of the sensor node. The placement of the nodes follows a similar approach as suggested in sec. III-A. The attacker model considered for the current investigation is explicated in the following section.

IV. ATTACKER MODEL

The local and passive attacker models are considered in this study. In this paradigm, the attacker just listens on the wireless communication links and tries to identify the originating source of the data packets by backtracking rather than launching active assaults like node compromise or packet content alteration. Furthermore, because the BS is the final destination of all packets, it is presumed that the attacker begins its backtracking effort from the BS by guessing the path of packet propagation using a device with spectrum sensing and analyzing capacity with sufficient storage and processing capability from the received packet at each transmission. As per the model, the attacker thus moves one hop in the direction of packet arrival towards the source of information and away from the BS. Furthermore, the devices of the attacker are assumed to have the same sensing radius as the sensor nodes. The SLP routing techniques considered for the comparative study are briefly described in the ensuing section.

V. ROUTING TECHNIQUES

In this section, we will briefly describe the three existing routing protocols namely, Phantom Routing Based Pure Random Walk (PRPRW), Sector Based Random Walk Routing (SRR), and Phantom Routing Based L-Path Pure Random Walk (PRLPRW) that will be used for comparing the impact of network shape and size on the performance of an SLP protocol.

A. SECTOR-BASED RANDOM WALK ROUTING

In the SRR technique, the authors have considered a circular deployment model for studying the SLP issue in habitat monitoring. In this case, the network is divided into imaginary annular rings, similar to the one shown in Fig 2, and further into logical sectors. This type of circular deployment was initially seen in [25] for SLP-related work and in [23] for general routing work. Initially, the packets are sent in the backward direction for two hops so that the packets are moving away from the source and the BS. At the conclusion of the backward walk, the packets are either relayed in the clockwise or anti-clockwise directions for a certain walk length. This walk length is random in nature and is determined based on a probability criterion. The immediate adjacent sector to the SN has the least probability while the one after that has the highest. The subsequent sectors have a decreasing probability of selection. This distribution is symmetric in nature for the northern and southern hemispheres of the conceptual circular deployment network and ensures packet transmission away from the SN while also avoiding unnecessary propagation till the network boundary. This is done to minimize the delay and energy consumption issues. Finally, the packets are sent to the BS in a forward random walk.

B. PHANTOM ROUTING BASED PURE RANDOM WALK

PRPRW, originally proposed for square deployment [16] and works in two phases: i) the first one is the initialization phase during which each node in the network discovers its one-hop neighbor and its distance to the BS, and ii) the second one is the operation phase: in this phase, once the asset/event is detected by a sensor node it starts sending packets to the BS using the random walk method. Initially, the source chooses a neighbor node randomly from its neighbor list and relays the packet to that node. Before relaying the packet, the SN initializes the packet with a time-to-live (TTL) parameter, which is a predefined hop count value. The neighbor node that receives the packet decrements this TTL value by one, chooses a neighbor randomly from its neighbor list, and relays this packet to that neighbor. This process repeats till the TTL value becomes zero, post which the packet is relayed to the BS using the shortest path routing (SPR) approach. The node that has a packet with a TTL value equal to zero is termed as a *phantom node*. The phantom node chooses a neighbor node that is closer to the BS compared to its other neighbors and relays the packets to that neighbor. Thus achieving the shortest path routing up to the BS.

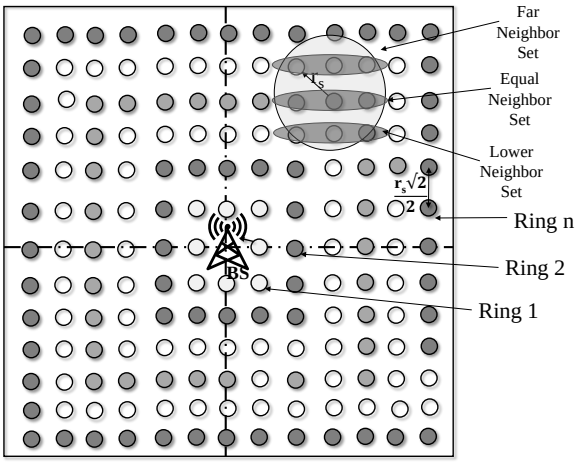


FIGURE 1: Square Deployment Area

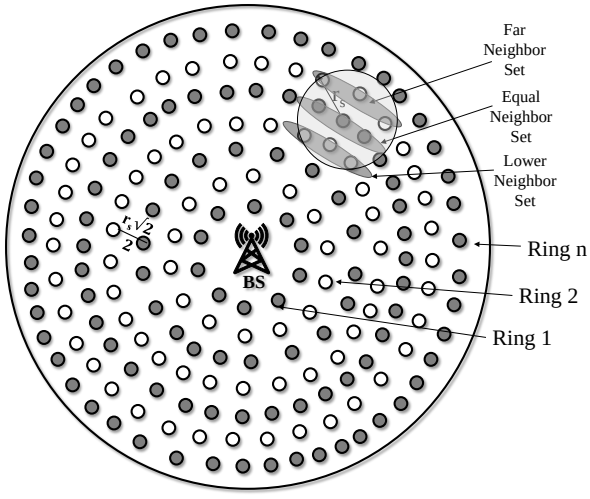


FIGURE 2: Circular Deployment Area

TABLE 1: Summary of Simulation Scenarios.

Cases	Area	Node Density (no. of nodes)	Radio range r_s
`Case I`	Same (4500000 square units)	Different (Square: 1849 nodes and Circular: 1746 nodes)	Constant $r_s = 72$ units
`Case II`	Different (Square: 4410000 square units and Circular: 3463606 square units)	Approx. Same (Square: 1849 and Circle: 1462 nodes)	Constant ($r_s = 72$ units)
`Case III`	Different (Square: 3422500 square units and Circular: 3141593 square units)	Approx. Same (Square: 1369 and Circle: 1330 nodes)	72, 77, 82, 87, 92, 97, 102, 107 and 112

C. PHANTOM ROUTING BASED L-PATH PURE RANDOM WALK

PRLPRW, initially introduced for square-shaped network [22] works similar to PRPRW but has an additional phase. During the first phase, the initialization of the network takes place in a way similar to the PRPRW technique. In the second phase, the packet moves in the L-path direction from the SN. Initially, the packet is sent either vertically up or down for certain hops and then horizontally left or right for another set of hops till a total of L hops is completed. This phase is intended to drive the packets away from the SN by introducing biases in the random walk. Finally, in the third phase, the packets are sent to the BS using the SPR approach as in the previous technique.

VI. SIMULATION AND EVALUATION METRICS

In this section, we explicate the various simulation scenarios used for assessing the impact of network shape, size, and the radio range of sensors on the performance of the examined routing protocols. The section concludes with a description of the different performance metrics used for the assessment.

A. SIMULATION SCENARIOS

In the simulation, four different positions of the SN at distances of 250, 550, 900, and 1050 units are considered. At each position using one of the three protocols described in Section V, 50 trials are conducted and during each trial, 500 packets are sent from the SN. The entire process is repeated for all the investigated routing protocols. While each trial initiates with the SN sending packets to the BS, it ends when one of the following occurs

- Attacker finds the SN.
- 500 packets are delivered to the BS.

The simulation is implemented using Python programming. Three different cases as summarized in Table 1 are used to study the impact of network shape, size, and sensor radio range on the performance of the considered routing protocol. The details of each of the cases are provided as follows:

- `Case I`: In this case, the network area and the sensors' radio range of both the network models (square and circular) is kept fixed at 4.5 million square units and 72 units respectively. An offset distance of 50 units is maintained for both deployment models. Furthermore, due to topological deviation, the square and circular deployments accommodate 1849 and 1746 nodes respectively within the same area thereby having different node densities. Node density in the current context is defined as the area occupied by a single node.
- `Case II`: In contrast to the previous scenario, in this case, the node density and the radio range of the nodes are held fixed at 1 node in approximately 2400 square units of area and 72 units respectively for both network

topologies. In an effort to maintain the same node density, however, the total network area between the deployment models becomes different – 4.41 million square units (square) and 3.46 million square units (circle). The same offset distance as before is maintained in this case as well.

- `Case III`: The radio range of the sensor nodes is varied from 72 units to 112 units in steps of 5 units for both circular and square models in this case. In addition, an effort is also made to keep the area ($\sim$ 3.4 million square units for square vs $\sim$ 3.1 million square units for circle) and the node density (1 node in 2500 square units of area for square vs 1 node in 2400 square units of area for circle) as similar as possible, and the offset distance is set to 50 units for both the deployment models.

While `Case I` and `II` enable the assessment of the impact of network area and shape on the performance of the inspected routing protocols, `Case III` aids in analyzing the impact of sensor radio range and network shape on the performance of the same routing protocols. The evaluation metrics used in the performance assessment are next described.

B. EVALUATION METRICS

Here, we list and briefly explain various metrics that have been considered in the literature [16], [23], [24], [38], [39] and used in this work to analyze and evaluate the performance of the SLP solutions.

1) Safety Period

The number of packets successfully delivered to the BS before the attacker discovers the SN is known as the *safety period*. Thus in these results, the unit of this parameter is *number of packets*.

2) Capture Ratio

The percentage of trials in which the attacker successfully identifies the SN before the end of the trial is called the *capture ratio*, or the *capture percentage*. That is capture ratio indicates the *attacker's success rate*. To cite an example, in these simulations 50 trials are conducted per SN position per protocol. Out of these 50 trials if the attacker discovers the SN in 20 trials then the *capture percentage* is 40%. Hence, the results for this parameter are expressed in *percentage*.

3) Entropy

The measure of average uncertainty in a set of data is generally defined as *entropy*. In the communication domain it is adopted to determine network traffic uncertainty. In the current context the unpredictability in routing formed by random walks is measured using *entropy* as given in [39]. The more random the walks are, the better the entropy which in turn implies a higher safety of the SN.

4) Delay

The average number of hops required by a data packet to reach the BS from its SN is termed as *delay*. In this context, the packet movement from one node to the next node is defined as a *hop* and in results, the parameter is measured in terms of *number of hops*

5) Energy Consumption

Processing information packets, as well as sending and receiving the packets, requires energy. In this context, only the energy expended during data transmission between nodes is taken into consideration in various literature as *energy consumption* and is measured in *Joules* in this work [38].

6) Network Lifetime

The time interval between the beginning of the network operation and the energy exhaustion of one of the nodes is defined as *network lifetime*. In this case, we send 1000 packets per trial and measure the metric in terms of the number of packets successfully sent in one trial before one of the nodes exhausts its energy.

In the subsequent section, we next present the performance of the considered routing protocols and their corresponding interpretation using the described simulation scenarios and the evaluation metrics presented in this section.

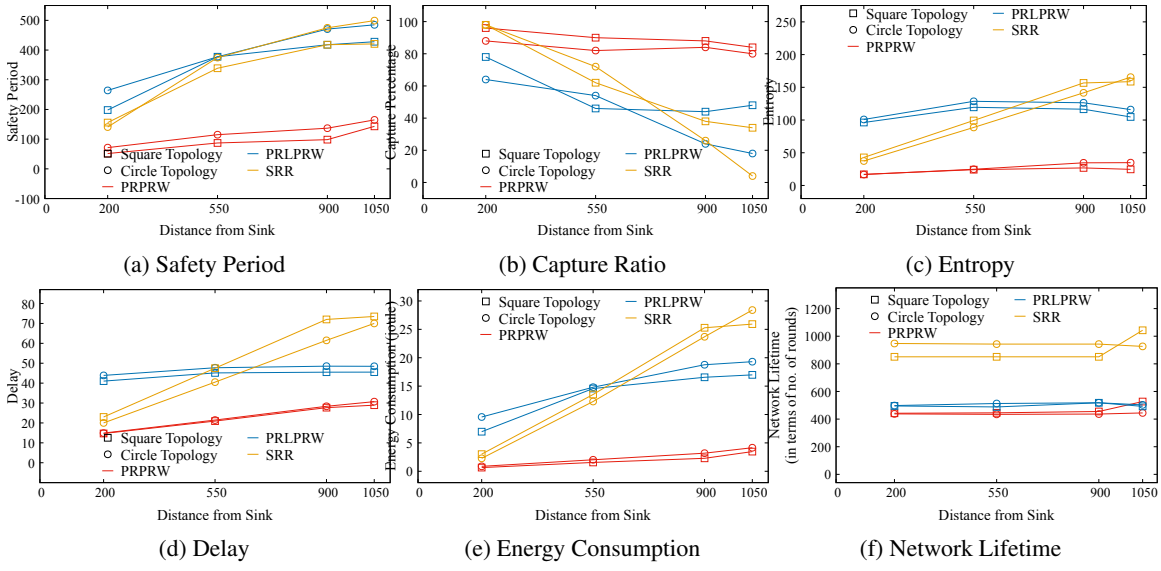
VII. RESULTS

In this section, we present and analyze the main findings of the work. In particular, we try to understand the effect of network shape (`Case I, II` and `III`), size (`Case I` and `II`) and radio range of the sensor nodes (`Case III`) on the performance of the existing SLP solutions (used as test cases).

A. `CASE I`: RESULTS AND INSIGHTS

Fig. 3 presents the variation in performance of the considered routing protocols in `Case I` as described in Section VI-A and summarized in Table 1. In the graphical depictions, the red, flaxen, and blue plots are corresponding to PRPRW, SRR, and PRLPRW protocols respectively while the square and circular markers represent the namesake models.

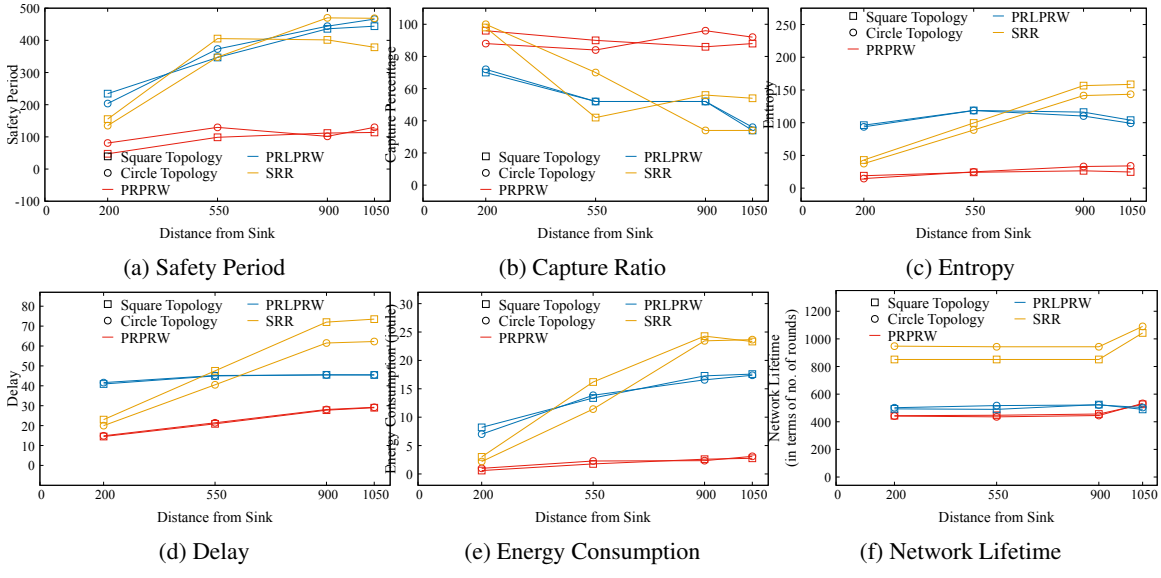
In this context the performance of the SLP techniques in `Case I` in terms of safety period (Fig. 3a), capture ratio (Fig. 3b), entropy (Fig. 3c), delay (Fig. 3d), energy consumption (Fig. 3e) and network lifetime (Fig. 3f) is presented. It is observed that irrespective of the technique the safety period for the circular deployment model is better than the square deployment model over all the distances between the SN and the BS. In addition, while SRR and PRLPRW have comparable performance in terms of the number of packets successfully delivered by the SN before being discovered, PRPRW lacks quite a bit. In regards to capture ratio, the value is mostly greater for square deployment in comparison to circular deployment, the only exception being PRLPRW and SRR when the separation between BS and SN is 550 units. As a lower value of capture ratio indicates better privacy, circular deployment would be preferable for most scenarios irrespective of the technique. On the other hand, the value of entropy for square deployment is smaller for



(a) Safety Period (b) Capture Ratio (c) Entropy

(d) Delay (e) Energy Consumption (f) Network Lifetime

FIGURE 3: Impact of deployment models on the performance of explored routing protocols in `Case I`.



(a) Safety Period (b) Capture Ratio (c) Entropy

(d) Delay (e) Energy Consumption (f) Network Lifetime

FIGURE 4: Impact of deployment models and distance from BS on the performance of explored routing protocols in `Case II`.

PRPRW and PRLPRW but larger for SRR. Since a larger value of entropy suggests greater randomness and hence higher security, circular networks should be preferred for the former two protocols and square model for SRR.

The plots for the delay metric suggest a higher delay for circular network models when using PRPRW and PRLPRW and for square models when using SRR. As for energy consumption, the circular deployment seems to be conservative for all the SN positions regardless of the routing protocol except when it is near the edge of the network. In a similar vein, the network lifetime has a higher value for circular networks for all SN positions and protocols except when the farthest SN position is concerned. The result makes sense since a larger energy consumption for the farthest SN position naturally translates into one of the nodes exhausting its energy quicker and hence lowering the network lifetime.

1) Insight and Conclusion

The results in Fig. 3 indicate that network shape and size indeed affect the performance of a routing protocol. While results of most of the privacy metrics (safety period, and capture ratio) indicate that circular deployment is more secure irrespective of the routing protocol there is a deviation in the results of entropy where square deployment suggests more privacy only in the case of SRR. The probable reason for such a trend in SRR can be attributed to the sectoral approach in

TABLE 2: Percentage deviation of performance metrics of different protocols in `Case I`.

Metrics	% Deviation in PRPRW	% Deviation in PRLPRW	% Deviation in SRR
Safety Period	28.44	12.35	11.69
Capture Ratio	-06.70	-25.92	-13.79
Entropy	19.86	7.98	-05.32
Delay	3.43	6.43	-11.11
Energy Consumption	28.54	13.34	-01.60
Network Lifetime	-06.22	01.93	04.55

TABLE 3: Percentage deviation of performance metrics of different protocols in `Case II`.

Metrics	% Deviation in PRPRW	% Deviation in PRLPRW	% Deviation in SRR
Safety Period	18.18	01.75	06.03
Capture Ratio	00.00	01.92	-04.80
Entropy	13.00	-03.08	-10.17
Delay	01.54	00.52	-17.25
Energy Consumption	12.83	-02.87	-09.11
Network Lifetime	-00.98	02.31	09.11

the protocol to increase the randomness. In contrast to circular deployment, the sector size is going to be non-uniform in square deployment in SRR, thereby accommodating more nodes in the larger sectors which contributes to additional randomness and hence entropy. The same reason would justify the reverse trend in the delay results in SRR where square deployment suffers from a higher delay among the two network models in comparison to circular deployment for the other considered protocols. A higher delay generally translates to higher energy consumption and a lower network lifetime. Thus circular model in the case of PRPRW and PRLPRW consumes more energy and has a lower network lifetime while the same is true for the square model in SRR. However, there is an aberration in this trend when the SN is at a distance of 1050 units from the BS. The reason for such a deviation is hidden in the shape of the network where the radius of a circle of area 4500000 units is ~ 1196 units whereas the length of a side of a square of the same area is ~ 2121 units. Thus half of the side has length ~ 1060.5 units. Thus for square models the backward random walk remains short, thereby saving energy and hence improving network lifetime in comparison to circular models. The deviation in performance of the models is further quantified and summarized in Table 2. For safety period, entropy, and network lifetime, a positive value indicates the circular model would be a better choice from the perspective of the metric being considered whereas a negative value is indicative of the same for the rest of the metrics. The results and analysis for `Case II` are reported in the next subsection.

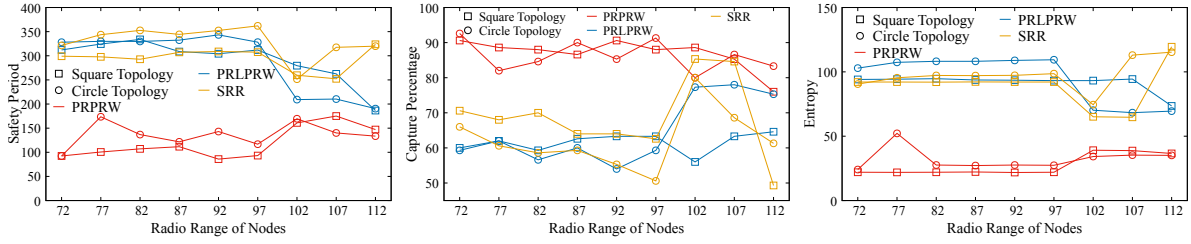
B. `CASE II`: RESULTS AND INSIGHTS

Similar to the depiction of results for `Case I` in Fig. 3, the performance variation of considered routing protocols for different network shapes in `Case II` is shown in Fig. 4. As before the red, flaxen, and blue plots in the depictions correspond to PRPRW, SRR, and PRLPRW respectively where the marker shape represents the corresponding network shape. It is prudent to mention in this context that in `Case II` as described in Table 1 the area of a square deployment is larger than the circular deployment. In this context, the variation of privacy parameters – safety period, capture ratio, and entropy is presented in Fig. 4a, 4b and 4c respectively. It is observed that a circular deployment with a smaller area but the same node density provides a better safety period on average irrespective of the routing protocol but the trend is not so obvious for the other two metrics. In this light, it is interesting to note that despite the difference in the area of the deployment shapes, the PRLPRW produces comparable performance in terms of capture ratio and entropy while PRPRW does the same in terms of entropy. On the other hand, due to the sectoral implementation in SRR, the randomness in packet propagation increases and the chances of the SN getting discovered by the adversary are less when it is closer to the BS in case of square deployment.

The non-privacy metrics such as delay, energy consumption, and network lifetime are shown in Fig. 4d, Fig. 4e, and Fig. 4f respectively. Clearly, SRR shows better performance with smaller delay, hence lesser energy consumption and subsequently larger network lifetime. This is for the case of a circular deployment model (having a smaller area when compared with the square deployment model). In the case of the other considered protocols, all the metrics have comparable variation for both the circular and square network shapes.

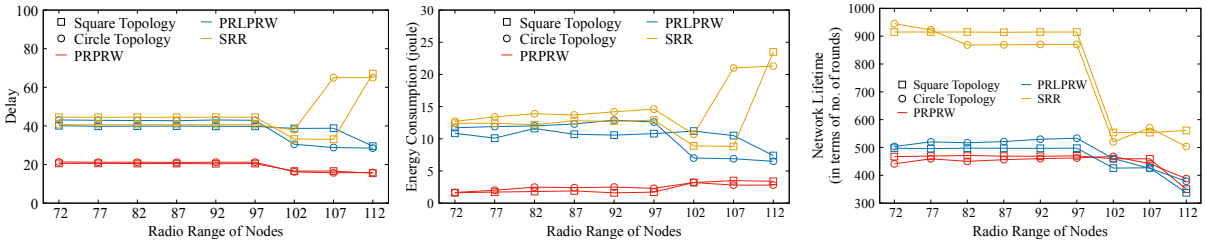
1) Insight and Conclusion

It can be seen from Fig. 4 that even when the node density and the radio range of nodes in a given deployment area are the same, the shape of the deployment area has an impact on the performance of a routing protocol. The extent of the impact of network shape is quantified and summarised in Table 3. As before in the case of safety period, entropy, and network lifetime a negative value indicates that the square-shaped network performs better in regard to the respective parameter being considered whereas a positive value for the rest of the metrics indicates the same. While circular deployment can provide better privacy in PRPRW, the square deployment seems to have a slight edge in regard to non-privacy metrics as clearly evident from the summary in Table 3. The result is quite surprising where a square deployment with a larger area and more nodes (to maintain the same node density) creates less uncertainty and safety period. The plausible reason for the same is not very apparent. The results for PRLPRW on the other hand make sense where the square deployment provides more privacy in terms of capture ratio and entropy simply from having more nodes. The non-privacy metric for PRLPRW thus favors the circular deployment as better privacy in this protocol is achieved by driving the packets away from the SN initially. An almost similar trend is observed in SRR, where the circular deployment should be preferred in terms of safety period and capture ratio and



(a) Safety Period (b) Capture Ratio (c) Entropy

FIGURE 5: Different Radio Range: Safety Period, Capture Ratio and Entropy in PRPRW, PRLPRW and SRR.



(a) Delay (b) Energy Consumption (c) Network Lifetime

FIGURE 6: Impact of deployment models and the radio range of nodes on the performance of considered routing protocols in `Case III`.

square deployment otherwise. The reason for the same can be attributed to the sectoral approach used in SRR as explained in Section VII-A1. Despite the same node density in `Case II`, the sector dimensions are still going to be non-uniform in square deployment and hence can accommodate a non-uniform number of nodes in the sectors explaining the current results. It also explains the results in SRR for the non-privacy metric where a similar trend is observed as in `Case I`.

C. `CASE III`: RESULTS AND INSIGHTS

The results for `Case III` are depicted in Fig. 6 where the color schemes and the markers used in the plots are the same as described in the previous two subsections. From Fig. 5a, 5b and 5c, it is evident that SRR when employed in a circular deployment provides better privacy in terms of the metrics considered. On the other hand, for PRPRW and PRLPRW the trend is not so binary. The entropy for the protocols is more when a circular deployment is employed with the radio range being less than $\sim$ 100 units post which the trend reverses. A similar trend is observed in regard to the safety period as well. However, in the context of the capture ratio, there is no clear trend.

In terms of non-privacy metrics, SRR implemented in a circular-shaped network certainly has an edge due to lower latency and higher network lifetime on average. It is to be noted that this trend is just the reverse of what we have seen in `Case I` and `II`. An interesting result in this context is the plot of energy consumption (Fig. 6a) where circular deployment consumes more energy in SRR despite having a lower average latency. In PRPRW the impact of radio range on the non-privacy metrics is minimal with square deployment doing slightly better on average in terms of energy consumption (Fig. 6a) and network lifetime (Fig. 6c). In the case of PRLPRW, the results are better in terms of delay and network lifetime with a circular deployment but a similar conclusion cannot be made in the case of energy consumption.

1) Insight and Inferences

A consistent trend with a variation of the radio range of nodes is observed in regard to the non-privacy metrics such as – delay, energy consumption, and network lifetime irrespective of the routing protocol. The radio range of nodes seems to have a noticeable influence on the metrics only when it exceeds $\sim$ 100 units. It is also interesting to note in this regard that increasing the radio range of nodes beyond a certain limit gives no advantage in regard to latency and energy efficiency irrespective of the protocol and network shape. In the context of privacy metrics – safety period, capture ratio, and entropy a change is observed once the radio range exceeds $\sim$ 100 units. However, the change is favorable only in the case of PRPRW where the safety period and entropy improve beyond the limit. In the case of PRLPRW, the SLP reduces beyond the limit in which a circular deployment suffers more. Finally, for SRR worst performance from a privacy perspective is observed when the radio range of nodes is set to 102 units. In this case, however, the square deployment suffers more than the circular one.

Table 4 summarizes the observed sensitivity of the studied SLP protocols to deployment geometry and radio range, highlighting their suitability for different deployment scenarios.

TABLE 4: Sensitivity of Random Walk–Based SLP Protocols to Deployment Geometry and Radio Range

Protocol	Geometry Sensitivity	Radio Sensitivity	Best Use Case
PRPRW	Low	High	Circular deployments with moderate radio range
PRLPRW	Moderate	High	Circular deployment with larger radio range
SRR	High	Moderate	Square deployment with small to moderate radio range

TABLE 5: Comparative Performance Trends of Random Walk–Based SLP Protocols under Square and Circular Deployments

Protocol	Square Deployment	Circular Deployment	Dominant Sensitivity
PRPRW	Moderate performance; improves with larger radio range	Better average privacy and efficiency	Radio range
PRLPRW	Moderate performance; energy-efficient at higher ranges	Better privacy–efficiency trade-off	Radio range
SRR	Strong performance across all metrics	Reduced effectiveness compared to square	Deployment geometry

Table 5 summarizes the observed performance trends of the studied SLP protocols under square and circular deployments, highlighting the distinct geometry-dependent behavior of sectoral and non-sectoral routing approaches.

VIII. DESIGN INSIGHTS AND DEPLOYMENT GUIDELINES

Based on our research findings, we provide the following guidelines for random walk based SLP protocols for WSN.

- Geometry-aware protocol selection: Sector-based protocols such as SRR benefit from square deployments due to non-uniform sector population, whereas non-sector-based protocols achieve better privacy in circular deployments.
- Radio-range optimization: Increasing radio range beyond a threshold (100 units) yields diminishing or adverse effects on privacy and energy efficiency, indicating that aggressive range scaling is not always beneficial.
- Trade-off awareness: Circular deployments generally improve safety period and energy efficiency but may reduce entropy under sector-based routing, highlighting a privacy–efficiency trade-off.
- Application-driven deployment: For terrain-constrained or wildlife-monitoring deployments where circular coverage is natural, non-sector-based protocols are preferable; structured square deployments favor sector-based approaches.

IX. CONCLUSION

This work demonstrates that source location privacy performance in wireless sensor networks is an emergent property of routing logic, deployment geometry, and radio configuration, rather than protocol design alone. When network area or node density is held constant, sector-based routing exhibits superior privacy and efficiency performance in square deployments, whereas non-sector-based approaches benefit more from circular deployments. In contrast, when radio range is varied under fixed area and density, performance improvements are observed only beyond a threshold range, revealing a saturation effect. Under such conditions, non-sector-based protocols achieve better performance in square deployments, while sector-based routing favors circular geometries. These results collectively indicate that SLP performance is strongly dependent on deployment architecture and radio configuration, underscoring the importance of geometry-aware protocol selection. As future work, the investigation can be extended to more realistic random and strategic deployment models and stronger adversary settings to further validate the generality of the observed trends.

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